Jianwen Luo

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EDUCATION

Nanjing University, School of Integrated Circuits, Nanjing, China Sep. 2024 – Expected Jun. 2028
B.S. Candidate in Integrated Circuit Design and Integrated System

- **GPA:** 4.71/5.00 | **Major Rank:** 2/168 | **English:** CET-4: 636, CET-6: 584
- **Relevant Coursework:** Microelectronics, Semiconductor Physics, Integrated-Circuit Devices, Analog & Digital Integrated Circuits, Signals and Systems

RESEARCH EXPERIENCE

ISCL Lab, Nanjing University, Nanjing, China Jul. 2026 – Present
High-Speed Interconnect Transceiver with Dielectric Waveguide

- Designed a D-band sub-harmonic mixer module in TSMC 28nm CMOS process, executing the full IC design flow including schematic design, physical layout, and tape-out verification using Cadence Virtuoso.
- Conducted 3D electromagnetic (EM) modeling and co-simulation in Ansys HFSS for millimeter-wave passive components (inductors and transmission lines) to accurately extract and mitigate high-frequency layout parasitics.
- Optimized the broadband impedance matching networks and enhanced port isolation by refining the symmetric layout of the mixing core, effectively compensating for severe parasitic degradation at frequencies above 100 GHz.

National College Student Innovation Training Program, Nanjing, China Nov. 2025 – Present
Project Lead (National-Level) | Multi-Modal Flexible Tactile Sensor for Robotic Material Recognition

- Spearhead overall project architecture and technical roadmap for a robotic multi-modal tactile sensing system.
- Design and validate mixed-signal peripheral readout circuits for high-precision material conductivity measurement.
- Perform finite-element analysis (FEA) in Abaqus and structural modeling in AutoCAD to optimize the modulus-sensing unit, completing sensor fabrication and experimental characterization.

National College Student Innovation Training Program, Nanjing, China Nov. 2025 – Present
Co-Lead (Provincial-Level) | Reconfigurable Computing Platform for UAV Edge-Side Intelligent Vision

- Construct experimental flight testbeds and execute field data acquisition to build, annotate, and filter multi-scenario UAV vision datasets.
- Optimize YOLO object-detection architectures and deploy quantized models onto reconfigurable edge-computing hardware for real-time inference.

National College Student Innovation Training Program, Nanjing, China Nov. 2024 – Mar. 2026
Team Member (University-Level) | Trade-offs and Evolution in Fundamental AI Research

- Conducted large-scale literature database mining, implementing automated data cleaning, pre-processing, and quantitative statistical analysis using Python.
- Evaluated project outcomes and secured an excellent completion rating from the university review committee.

HONORS & AWARDS

Scholarships

- **National Scholarship**, Ministry of Education of China 2026
- **Guo-Xie Birong Special Scholarship**, Nanjing University 2025

Competitions & Honors

- **Special Championship Award**, The 2nd Low-Altitude Industry Innovation Contest 2026
- **3rd Prize**, The 2nd Low-Altitude Industry Innovation Contest 2026
- **Honorable Mention (H Award)**, Mathematical Contest in Modeling (MCM/ICM) 2026
- **1st Prize**, NJU Electronic Process Innovation and Instrument Measurement Competition (RIGOL Cup) 2025

TECHNICAL SKILLS

Languages & HDL: C/C++, Verilog-HDL, Python, MATLAB, LaTeX
EDA & Engineering Tools: Cadence Virtuoso, Abaqus, AutoCAD, Multisim, JLC-EDA, VS Code, Arduino